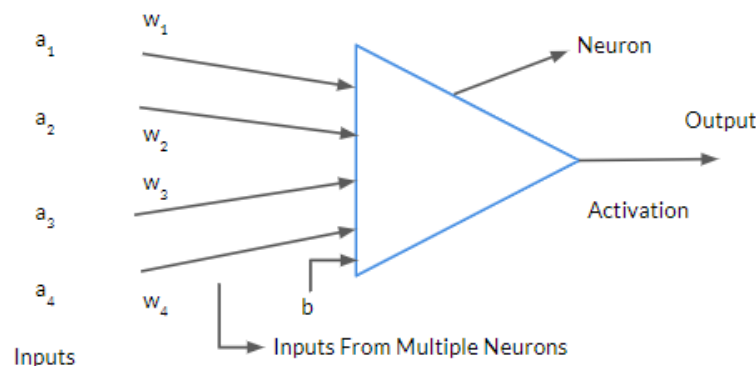


Summary

Let's take a quick look at what you have learnt in this session.

Some important points can be summarised as follows:

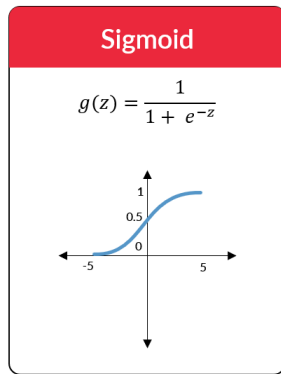
1. Firstly, you understood the limitations of preliminary machine learning and how deep learning can be used to build complex models.
2. Next, you saw how the architecture of ANNs draws inspiration from the human brain.
3. You also learnt about the basic functioning of a perceptron.
4. Further, you learnt about the basic building block of ANNs: Neurons. The structure of an artificial neuron is shown below.



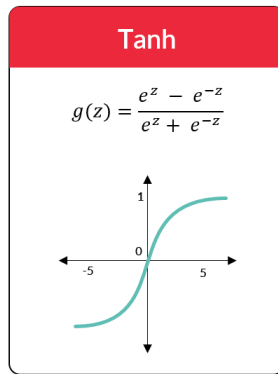
Here, 'a' represents the inputs, 'w' represents the weights associated with the inputs, and 'b' represents the bias of the neuron.

5. You then learnt about the architecture of ANNs, including the topology, the parameters (weights and biases) on which the neural network is trained and the hyperparameters.
6. ANNs only take numerical inputs. Hence, you need to convert all types of data into a numeric format so that neural networks can process it.
7. Next, you were introduced to the most common activation functions such as sigmoid, ReLU, Leaky ReLU and tanh, which are shown below.

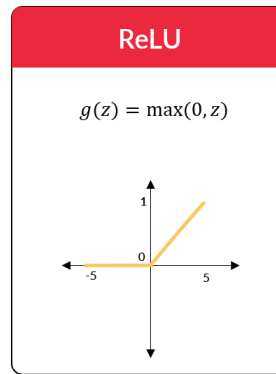




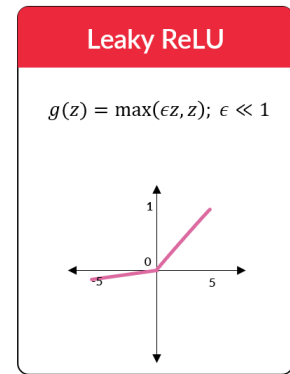
Output values bound between **0 and 1**.
Outputs not zero-centered.



Output values bound between **-1 and 1**.
Outputs are **zero-centered**



Computationally efficient: allows the network to converge very quickly.



Allows learning even when the value is negative.

8. Some simplifying assumptions in the architecture of ANNs are as follows.

1. The neurons in an ANN are **arranged in layers**, and these layers are arranged **sequentially**.
2. The neurons within the same layer **do not interact** with each other.
3. The inputs are fed into the network through the **input layer**, and the outputs are sent out from the **output layer**.
4. Neurons in **consecutive layers** are **densely connected**, i.e., all neurons in layer l are connected to all neurons in layer $l+1$.
5. Every neuron in the neural network has a **bias** value associated with it, and each interconnection has a **weight** associated with it.
6. All neurons in a particular hidden layer use the **same activation function**.

9. Finally, you fixed the following notations:

1. W represents the weight matrix.
2. b stands for bias.
3. x represents input.
4. y represents the ground truth label.
5. p represents the probability vector of the predicted output for the classification problem.
6. h represents the output of the hidden layers, and h^L represents the output prediction for the regression problem.
7. z represents the cumulative input fed into each neuron of a layer.
8. The superscript represents the layer number.
9. The subscript represents the index of each individual neuron in a layer.

in the next segment, you will attempt some graded questions to test your understanding of the topics covered in this session. All the best!

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Assumptions for Simplifying Neural Network

Graded Questions-Graded Quiz

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